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Immutable COVID-19 Vaccination Certificate using Blockchain

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Abstract

The COVID-19 pandemic has presented numerous challenges around the world, including lockdowns, remote working and studying, and travel restrictions imposed by governments to curb the spread of the virus. As vaccines have become more widely available, restrictions have begun to ease for those vaccinated. Many countries use paper-based COVID-19 vaccination certificates to prove vaccination status. However, traditional certificates are vulnerable to forgery and counterfeiting. We developed a blockchain-based system where vaccination certificates are stored and accessed via smart contracts on an Ethereum blockchain. The certificates are stored in a tamper-proof, decentralized manner, ensuring secure verification of vaccination status. This research project successfully designed and implemented a system that securely stores and verifies vaccination certificates using blockchain, demonstrating the benefits of this approach. According to our knowledge, this is the first kind of research initiative in Brunei to develop a blockchain-based immutable COVID-19 Vaccination certificate. Despite its advantages, blockchain still has its flaws, particularly in scalability and adoption, which should be considered for further optimization of the proposed system.

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1. Introduction

The COVID-19 pandemic significantly impacted the global healthcare system, prompting widespread restrictions

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to curb the virus's spread [1]. Examples of these restrictions include lockdowns, social distancing measures, travel bans, and closures of non-essential businesses. As vaccination efforts increased, these restrictions gradually lifted. COVID-19 vaccination certificates are now being used worldwide to facilitate the reopening of travel and hospitality sectors [2]. This research examines the limitations of traditional paper-based vaccination certificates and proposes an immutable digital vaccination certificate solution using blockchain technology.

1.1 Problem Statement

Traditional paper-based vaccination certificates are vulnerable to manipulation, resulting in fake COVID-19 vaccinations and certificates [3, 4]. This issue presents public health risks, as individuals with the virus may falsify their vaccination status and spread the disease [5]. Counterfeit certificates also cause border control issues, delaying immigration and burdening authorities [6, 7]. The growth of vaccination certificates also fuels the market for counterfeit certificates, available on the dark web and messaging apps [4]. Buyers of fake certificates often spread misinformation and reject vaccination based on conspiracy theories [4]. Developing a secure, tamper-proof vaccination certificate system is essential to safeguard public health, resume activities safely, and restore authentication trust, necessitating innovative technology for reliable and secure verification of vaccination status.

1.2 Research Objectives

This research project aims to investigate the security and ease of use issues associated with traditional paper-based COVID-19 vaccination certificates and explore digital-based certificates using blockchain. The objectives include analysing the use of blockchain for certificate issuance, storage, and verification, designing and developing a decentralized system using Ethereum blockchain and smart contract as a secure method for storing and verifying patients' COVID-19 vaccination details. Lastly, to create a tamper-proof and immutable record of vaccination status using blockchain's features.

2. Background Study

Blockchain is a groundbreaking technology that has the potential to revolutionize various industries. It is a decentralized, distributed ledger technology that securely records transactions across a network of computers [8, 9]. This technology enables secure, transparent, and tamper-resistant data storage using advanced cryptographic techniques and consensus algorithms [10]. Blockchain technology can greatly improve the current vaccination certificate system as it addresses issues like forgery, counterfeiting, and lack of verifiability that plague traditional paper-based certificates [11]. A blockchain-based COVID-19 vaccination certificate called, CoviChain [1] framework is proposed by Bradish et al., based on biometrics, Locality Sensitive Hashing (LSH) and blockchain. Another blockchain-based digital vaccine passport is developed using Ethereum, and Solidity with a web interface. [12]. Nabil et al. developed a COVID-19 vaccination registration and monitoring using blockchain with a QR code [13]. The key properties of blockchain are decentralization, immutability, transparency, and distributed consensus. Blockchain enhances data security, traceability and authenticity that can safely store sensitive data like vaccination records and strengthen digital vaccination certificates [12, 14]. A research study on the possibility of using blockchain and smart contract-based digital certificate was carried out [15].

2.1 Key Properties of Blockchain

Table 1 shows the key properties of blockchain, which contribute to its potential for widespread application and adoption. These properties make blockchain a promising candidate for various use cases, such as supply chain management, financial transactions, digital identity verification, and many others. By leveraging the unique characteristics of blockchain, businesses and organizations can develop solutions that offer increased security, transparency, and efficiency.

Table 1. Properties of Blockchain.

Properties	Description
Decentralized	No central authority controls the system, reducing the risk of single points of failure.
Distributed	Data is stored across a network of computers, leading to increased security and redundancy.
Secure	Cryptographic techniques, such as public-key cryptography and SHA-256, protect data from unauthorized access and tampering.
Transparent	Transactions are publicly visible and verifiable, increasing trust and accountability.
Tamper-resistant	Consensus algorithms, such as Proof of Work (PoW) and Proof of Stake (PoS), make it difficult to alter data, ensuring the integrity of stored records.

2.2 Traditional vs. Blockchain-based COVID-19 Vaccination Certificates

As shown in Table 2 below, blockchain-based COVID-19 vaccination certificates are more secure, efficient, and private compared to traditional paper-based certificates. Blockchain-based certificates are tamper-proof due to cryptographic hash functions [16] and enable automatic verification through smart contracts, making the process more efficient than the manual verification required for paper certificates [16]. Blockchain-based certificates are also more accessible as they are digital, unlike paper certificates which can be easily lost or damaged [16]. Finally, blockchain-based certificates keep personal information private through encryption, addressing privacy concerns with paper certificates which display personal information openly [16].

Table 2. Traditional vs. Blockchain-based COVID-19 Vaccination Certificates.

Properties	Traditional	Blockchain-based
Security	Vulnerable to manipulation and fraud	Secure and tamper-proof due to cryptographic hash functions
Verification	Manual verification by healthcare personnel	Automatic verification using smart contracts
Accessibility	Physical certificates that can be easily lost or damaged	Accessible through digital platforms and devices
Privacy	Personal information is stored on the certificate	Personal information is kept private and secure using encryption

2.3 Key Advantages of Using Blockchain for Vaccination Certificate Management

Table 3 below shows the advantages of using blockchain for the management of vaccination certificates. Blockchain offers numerous advantages for vaccination records and verification. It provides fraud prevention by ensuring the authenticity of records, enhances data protection against breaches, streamlines communication and data accuracy, prevents duplicate vaccinations, and enables efficient access control through digital identity creation. These benefits contribute to increased trust, improved data integrity, and enhanced safety in public spaces.

Table 3. Advantages of Using Blockchain for Vaccination Certificate Management.

Advantages	Description
Fraud Prevention	Immutable blockchain data prevents forged or falsified vaccination records, ensuring authenticity
Data Breach Protection	Blockchain's security features safeguard sensitive personal information, enhancing data protection
Clear Communication & Data Accuracy	Centralized digital records streamline vaccination verification, preventing miscommunication and data loss

Prevents duplicate vaccinations	Unique IDs and tracking prevent unnecessary multiple vaccinations and ensure accurate histories
Efficient access control	Blockchain certificates enable digital identity creation, enhancing safety in public spaces through seamless access management

2.4 Existing Implementation of Blockchain for COVID-19 Vaccination Certificate

As countries accept fully vaccinated travellers, managing travel documents has become complex due to varying vaccination documentation and quarantine requirements [2, 12]. To tackle this issue, blockchain has been employed to create credible vaccination documentation, such as the International Air Transport Association (IATA) Travel Pass, IBM Digital Health Pass, and Singapore's HealthCerts [2]. These systems ensure secure, verifiable, and decentralized data management, allowing users to own their data, and fostering trust [2]. The IATA Travel Pass, launched in December 2021, is a smartphone application that stores COVID-19 test results, vaccination records, and health status information [17]. Similarly, IBM Watson Health developed the IBM Digital Health Pass, which also utilizes blockchain technology for secure data sharing among various entities [2, 14]. Singapore's GovTech Agency introduced HealthCerts in 2021, a system built on their OpenAttestation open-source framework, enabling the creation of cryptographically trustworthy digital COVID-19 vaccination certificates [18, 19].

3. System Design

This report explores the development of an alternative to traditional paper-based vaccination certificates using the Ethereum blockchain. A blockchain-based COVID-19 vaccination certificate system would enable vaccination centers to issue digital certificates containing personal information, vaccination doses, vaccine type, and dates. Individuals could share these certificates with employers or government officials, who could then verify the certificate's authenticity by cross-referencing it with the blockchain to ensure it has not been tampered with. The project aims to create a functional prototype of a blockchain-based COVID-19 vaccination certificate using smart contracts. In response to current restrictions, organizations and businesses are increasingly requiring proof of vaccination, making this technology relevant for schools, stores, and airports as well. The system must be tamper-proof, transparent, easily integrated, efficient, and secure against potential cyber threats.

3.1 Technologies Used

Ethereum is a decentralized, open-source platform supporting smart contracts and decentralized applications (dApps), with Ether (ETH) as its cryptocurrency [12]. Ethereum was introduced by Vitalik Buterin in 2015, with the key feature being self-executing smart contracts, which automatically execute contract requirements once certain conditions are met [12, 20]. These contracts, written in Solidity language [21], can be utilized in various sectors like finance, logistics, and property transactions, promoting efficiency, safety, and transparency without intermediaries [22, 23].

Solidity, developed by the Ethereum Network Team, is a programming language specifically for writing smart contracts on the Ethereum blockchain, with a syntax resembling JavaScript and support for Ethereum Virtual Machine (EVM) capabilities [21]. Solidity Remix is a browser-based integrated development environment (IDE) that enables developers to write, test, and deploy smart contracts using Solidity, offering tools such as a code editor, debugger, virtual machine, and deployment tool without the need for a live Ethereum network [24].

3.2 System Overview

The system overview is shown below in Fig. 1 which utilizes the Ethereum blockchain and smart contracts to store and validate patients' vaccination records. First, the smart contract will be deployed by the owner (Ministry of Health) on the Ethereum blockchain. Then, the owner will add a center into the blockchain, which can be vaccination centers or other centers like airports, schools, offices etc. Patient information will then be added to the blockchain by the

vaccination centers, followed by the generation of the vaccination certificates. The centers that do the verification process can then run the smart contract that verifies the certificate with the patient's latest information.

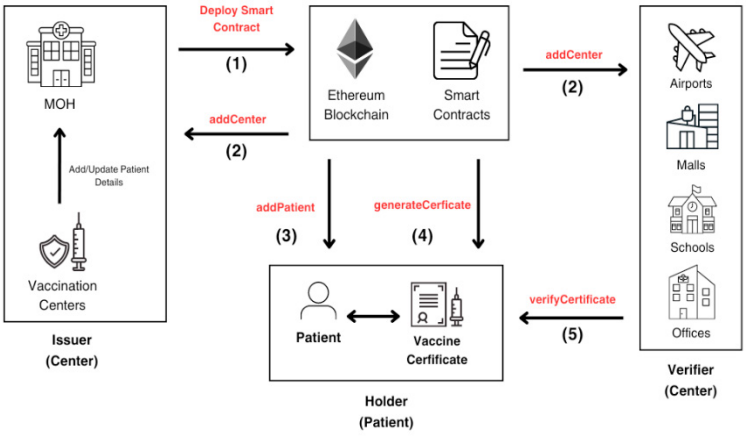


Fig. 1. System Overview

Table 4 below shows the four main entities in our system. The four main entities are, Smart Contract Owner, Issuer, Holder, and Verifier.

Table 4. System Entities with Examples.

Entities	Responsibilities	Example
Smart Contract Owner	Deploys and fully accesses code, manages rules, and oversees contract functions	Ministry of Health
Issuer	Provides and manages vaccination certificates, typically healthcare providers, or organizations. Adds new centers and patient information to the blockchain via smart contract.	Hospitals and Clinics
Holder	Receives the vaccine and digital vaccination certificate, provides personal information to the Issuer, and presents the certificate to the Verifier	Vaccinated Individuals
Verifier	Checks authenticity and validity of certificates using the smart contract's "verifyCertificate" function to confirm its legitimacy.	Airports, Workplaces, Malls, Schools

4. Implementation of the Smart Contract Code

The smart contract is named "VaccinationCertificate" and is divided into three sections: one for centers, one for patients, and one for the vaccination certificate.

4.1 Center Functions

The smart contract uses a "Center" struct as a blueprint for storing information about hospitals or clinics that issue vaccinations in the system. This struct comprises the center's Ethereum address, name, and location. Two mappings related to vaccination centers are used for efficient storage, retrieval, and quick verification of a center's existence. Table 5 below shows three primary functions that allow the contract owner to manage vaccination centers by adding, updating, and retrieving information as needed.

Table 5. Vaccination Center Management Functions

Advantages	Description
addCenter	Registers a new center, ensuring data integrity by checking non-empty inputs and verifying registration status
updateCenter	Updates an existing center's details, checking for non-empty inputs and verifying registration status
getCenter	Retrieves center details without altering the smart contract state, returning the center's address, name, and location or an error message if the center does not exist

4.2 Patient Functions

The Patient struct represents individuals who have received COVID-19 vaccinations and includes fields such as nationalId, name, dob, gender, passportNumber, telephoneNumber, dose, batch, brand, country, date, and certificateId. There are three primary functions (shown in Table 6 below) that manage patient information.

Table 6. Patient Information Management Functions

Smart Contract Functions	Purpose
addPatient	Adds new patient information to the blockchain
updatePatient	Updates existing patient information on the blockchain
getPatient	Retrieves specific patient information, checks patient existence and returns the patient's details

4.3 Certificate Management Functions

The Certificate struct represents a vaccination certificate with fields like certificateId, nationalId, patient's name, dob, gender, passport number, telephone number, dose number, batch number, vaccine brand, country, and vaccination date. The functions in Table 7 below allow for the creation, retrieval, and verification of vaccination certificates, providing a secure and organized way to manage vaccination records on the blockchain.

Table 7. Vaccination Certificate Management Functions

Smart Contract Functions	Purpose
generateCertificate	Creates a vaccination certificate for a patient using their national ID
getCertificateId	Retrieves the certificate ID of a patient using their national ID
getCertificate	Retrieves the details of a vaccination certificate using a given certificate ID

4.4 Access Control Functions

Table 8 below showcases the security measures implemented in the smart contract, where different entities have access to specific functions. This ensures that sensitive operations are only accessible to authorized parties, maintaining the overall security of the smart contract. The entities include the Smart Contract Owner, Issuer, Verifier, and Patient, and their access to functions is strictly controlled based on their roles and responsibilities.

Table 8. Access Control Functions by Entity Role

Properties	Entity			
	Owner	Issuer	Verifier	Patient
addCenter, updateCenter, getCenter	/	×	×	×
addPatient, updatePatient, getPatient	/	/	×	×

generateCertificate	/	/	×	×
getCertificateID	/	/	/	/
getCertificate	/	/	×	/

4.5 Input Validation Functions

Table 9 below presents a list of error outputs that may occur when executing specific functions in the smart contract. These error messages aim to provide clear and concise feedback regarding the nature of the issue encountered, such as unauthorized access, non-existent or duplicate centers, patients, or certificates, enabling users to address the respective problems accordingly.

Table 9. Functions Error Output and Description

Error Output	Description
Unauthorised Access!	Only contract owner (based on their Ethereum address) can perform this action
Center Doesn't Exist	Center Address not found in the Blockchain
Center Already Exist	Center is already added in the Blockchain
Patient Doesn't Exist	Patient not found in the Blockchain
Patient Already Exist	Patient already added in the Blockchain
Patient Already has a Certificate	Prevents duplication of certificates
Certificate Doesn't Exist	Certificate ID does not exist in the Blockchain

5. Smart Contract Outputs

In Solidity Remix, the log feature allows developers to view the output or events emitted by a smart contract during its execution. By defining events within the contract and emitting them during transactions, the log captures and displays these events, along with associated data, providing a convenient way to observe the contract's execution flow, debug, and monitor its behaviour. The following will show the outputs of a few functions of the smart contract.

5.1 Add center Function Output

Fig. 2 below shows part of the output log when a Center is successfully added to the blockchain. In this case, the "Ministry of Health" is the owner of the smart contract based on its Ethereum address.


```
"from": "0xd9145CCE52D386f254917e481eB44e9943F39138",
"topic": "0x5df55bb7845c91352bcc50bb9a8f81ff0d71ea7fd57f75e8d5c851ae95b2dc4f",
"event": "CenterAdded",
"args": {
        "0": "0x5B38Da6a701c568545dCfcB03FcB875f56beddC4",
        "1": "Ministry of Health",
        "2": "Bandar Seri Begawan",
        "centerAddress": "0x5B38Da6a701c568545dCfcB03FcB875f56beddC4",
        "name": "Ministry of Health",
        "location": "Bandar Seri Begawan"
```


Fig. 2. addCenter Function Output

5.2 Add Patient Function Output

Fig. 3 below shows part of the output log when a Patient's data is successfully added into the blockchain. The patient's personal information is displayed alongside their vaccination dosage information.


```
{
	"string nationalId": "01-09123",
	"string name": "Muqsit Jafari",
	"string dob": "11/11/1989",
	"string gender": "Male",
	"string passportNumber": "C0705123",
	"string telephoneNumber": "+6737103133",
	"uint256 dose": "1",
	"string batch": "CTMAV522",
	"string brand": "AstraZeneca Vaxzevria",
	"string country": "Brunei",
	"string date": "22 May 2021 10:25"
}

{}

[
	{
		"from": "0xd9145CCE52D386f254917e481eB44e9943F39138",
		"topic": "0xe731252065458a9665f0f7c97690db6520ce9702aa72154b112018b8b90a8ce9",
		"event": "PatientAdded",
		"args": {
			"0": "01-09123",
			"1": "Muqsit Jafari",
			"nationalId": "01-09123",
			"name": "Muqsit Jafari"
		}
	}
]

0 wei
```


Fig. 3. addPatient Function Output

5.3 Generate Certificate Function Output

One of the main functions of the smart contract is to generate a vaccination certificate based on a patient's information. Fig. 4 below shows part of the output log when a certificate is successfully generated using a patient's National ID, and a Certificate ID is issued in numerical order.


```
"from": "0xd9145CCE52D386f254917e481eB44e9943F39138",
"topic": "0x74cf4205f12d65af0b0296fa2029b0fecc8c11c3f081b8030fc8f7260a9287ef",
"event": "CertificateGenerated",
"args": {
	"0": "1",
	"1": "01-09123",
	"certificateId": "1",
	"nationalId": "01-09123"
```


Fig. 4. generateCertificate Function Output

5.4 Verify Certificate Function Output

Another main function of the smart contract is to verify the vaccination certificate with the patient's latest information. Fig. 5 below shows part of the output log when a certificate is successfully verified by comparing its hash value with the hash of the patient's information.

```
{
        "from": "0xd9145CCE52D386f254917e481eB44e9943F39138",
        "topic": "0x739766c917befd2312040cd8bdbcc45e456f5699d6e4466fb15dad5089660be0",
        "event": "CertificateVerified",
        "args": {
                "0": "1",
                "1": true,
                "certificateId": "1",
                "isVerified": true
```


Fig. 5. verifyCertificate Function Output

6. Challenges and Limitations

Developing a blockchain-based COVID-19 vaccination certificate system presented numerous challenges and limitations. Regulatory challenges, adoption concerns, and accessibility limitations are also potential issues for the blockchain-based COVID-19 vaccination certificate system. Addressing these issues may require collaboration with regulatory bodies, promoting public education and awareness campaigns about blockchain technology, and providing alternative methods of obtaining and presenting vaccination certificates for those without digital devices or internet access. For real-time implementation, scalability is a major concern, where alternative consensus mechanisms or layer-2 scaling solutions should be considered to improve transaction processing times.

7. Conclusion and Future Research

During the COVID-19 pandemic, the need for a secure alternative to paper-based vaccination certificates became apparent. This research project developed a blockchain-based system using Ethereum's smart contract features addressing security, efficiency, and transparency concerns associated with traditional vaccination certificates. The new system offers a more secure and reliable method for managing vaccination records compared to conventional paper-based certificates. The smart contract code included functions for registering vaccination centers, enrolling patients, generating certificates, and verifying them, with thorough error handling and security measures. Despite challenges such as the learning curve for blockchain technology, scalability concerns, and technology adoption, the project demonstrated the practical application of blockchain for managing vaccination certificates. Future work could focus on advanced privacy features, alternative blockchain platforms, and user-friendly interfaces for stakeholders. By addressing these aspects, the project can contribute to a more effective response to future health crises and promote a secure and efficient way of managing health records.

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